

SOP-CSE-200: Confined Space Entry Safety Procedure

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Section 1. Purpose and Scope

This procedure establishes the minimum requirements for safe entry into permit-required confined spaces, including but not limited to storage tanks, pressure vessels, pits, sumps, and sewers. It applies to all employees and contractors performing confined space entry work on site.

A confined space is classified as permit-required if it contains, or has the potential to contain, a hazardous atmosphere, a material that could engulf an entrant, an internal configuration that could trap or asphyxiate an entrant, or any other recognized serious safety or health hazard.

Section 2. Pre-Entry Requirements

Clause 2.1 — A Confined Space Entry Permit (Form HSE-007) shall be completed and authorized by the Area Supervisor before any entry occurs. The permit is valid for a single shift only and shall not be extended without re-verification of all pre-entry conditions.

Clause 2.2 — Atmospheric testing shall be performed prior to entry and shall confirm: oxygen concentration between 19.5 percent and 23.5 percent, flammable gas below 10 percent of the Lower Explosive Limit (LEL), and toxic gas concentrations (H₂S, CO) below the site-specific exposure limits listed in Appendix A.

Clause 2.3 — If atmospheric testing indicates conditions outside the acceptable range in Clause 2.2, entry is prohibited until the space has been purged, ventilated, and re-tested. A minimum of two consecutive passing readings, taken 5 minutes apart, is required before entry is authorized.

Clause 2.4 — All energy sources connected to the confined space (electrical, mechanical, hydraulic, pneumatic, or process piping) shall be isolated per SOP-LOTO-400 prior to entry.

Section 3. Entry Team Roles and Responsibilities

Step 3.1 — Every confined space entry requires a minimum entry team of three: one Entrant, one Attendant, and one Entry Supervisor. The Attendant shall remain outside the space at all times and shall never enter the space to attempt a rescue.

Step 3.2 — The Entry Supervisor is responsible for verifying that all permit conditions are met, authorizing the permit, and having the authority to cancel the permit if conditions change.

Step 3.3 — The Attendant shall maintain continuous communication with the Entrant (visual, voice, or signal line) and shall monitor atmospheric conditions continuously using a calibrated four-gas monitor.

Step 3.4 — All entry team members shall complete Confined Space Entry training within the last 12 months, documented in the site Training Management System.

Section 4. Continuous Atmospheric Monitoring

Clause 4.1 — Continuous atmospheric monitoring is mandatory for the duration of the entry. Monitor alarms shall be set at: oxygen below 19.5 percent or above 23.5 percent, LEL above 10 percent, H₂S above 10 ppm, and CO above 35 ppm.

Clause 4.2 — Upon any alarm activation, all entrants shall evacuate the space immediately. Re-entry is prohibited until the cause of the alarm has been identified and corrected, and fresh atmospheric testing confirms conditions are within acceptable limits per Section 2.

Clause 4.3 — Gas monitors shall be bump-tested daily before use and calibrated per manufacturer specification, with calibration records retained for 12 months.

Section 5. Rescue and Emergency Procedures

Step 5.1 — A rescue plan specific to the confined space shall be documented on the entry permit before work begins, including the method of retrieval (mechanical retrieval system for vertical entries exceeding 5 feet) and the location of the nearest rescue team.

Step 5.2 — Non-entry rescue using a retrieval line and mechanical device is the preferred method and shall be used whenever the physical configuration of the space allows it.

Step 5.3 — In the event of an emergency, the Attendant shall immediately notify emergency services and the site Control Room and shall not enter the space under any circumstances, consistent with Step 3.1.

Section 6. Permit Closure

Clause 6.1 — Upon completion of work, the Entry Supervisor shall verify all entrants have exited, confirm tools and equipment are removed, and formally close the permit.

Clause 6.2 — Closed permits shall be retained by the HSE Department for a minimum of 5 years and are subject to review during incident investigations and regulatory audits.